

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CE/IT)

Nov 2013

CE-210 MICROPROCESSOR & COMPUTER ARCHITECTURE

Date: 16.11.2013, Saturday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
- 5.

SECTION-I

Q-1 Answer the following questions. [11]

- [A] Explain terms: Program counter and Stack Pointer [2]
- [B] Explain terms: Machine Language, Assembly Language, High-level Language [3]
- [C] What is use of following pins of 8085: HOLD, ALE, INTR [3]
- [D] Compare memory mapped I/O and peripheral mapped I/O. [3]

Q-2 Answer the following questions. [12]

- [A] Write an assembly program for multiplication of two 8-bit numbers. [5]
- [B] The instruction code 0100 1111(4F) is stored in memory location 2005H. Explain the data flow and list the sequence of events when the instruction code is fetched by the MPU with necessary diagram. [7]

OR

- [B] How does 8085 perform internal data operations? How many registers it has to perform internal data operations? Write functionality of each register. [7]

Q-3 Answer the following questions.(Attempt any Two) [12]

- [A] Draw and explain, Demultiplexing of bus AD7-AD0 in 8085.
- [B] List the sequence of events that occurs when the 8085 CPU reads data from Memory.
- [C] Write down various addressing modes of 8085 with example.
- [D] Draw and explain timing diagram of POP instruction.

SECTION-II

Q-4 Answer the following questions. [11]

- [A] How does a microprocessor differentiate between data and instruction code? [2]
- [B] Explain Instruction: ADI, STA, CMA [3]
- [C] Compare 8086, 80286 and 80386. [3]
- [D] If the device connected on the system bus is slow, explain the support provided by 8085 to synchronise its speed with the device speed. [3]

Q-5 Answer the following questions. [12]

- [A] What is DMA? How is it interfaced with 8085? [5]
- [B] List the four categories of 8085 instructions that manipulate data. Explain instruction of each category. [7]

OR

- [B] How 8085 does identifies and executes Interrupt service routine? [7]

Q-6 Answer the following questions.(Attempt any Two)

[12]

- [A] Draw and explain interfacing between 8255A and 8085 microprocessor.
- [B] Explain the functional diagram of 8155 and explain its control word.
- [C] Specify memory map of the following diagram and explain significance of the don't care address line on memory addresses.

